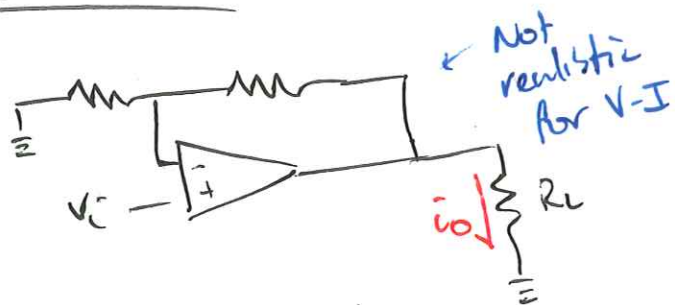
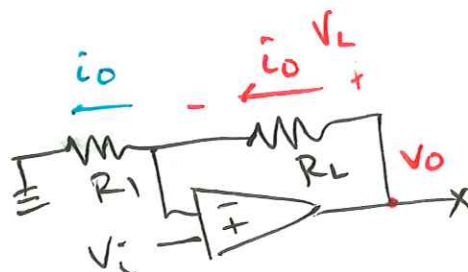


V-I converters



"Grounded ~~floating~~ load"

→ Amp only one terminal of load



"floating-load"

→ Amp "controls" both terminals of load.

Considering Right circuit:
Ideal Analysis:

$$\bar{i}_{RL} = \bar{i}_{R1} = \bar{i}_o$$

$$\frac{V_o - V_i}{R_L} = \frac{V_i}{R_1} = \bar{i}_o \rightarrow \text{looking for } \frac{\bar{i}_o}{V_i}$$

$$A_{Vi} = \frac{\bar{i}_o}{V_i} = \frac{1}{R_1}$$

Voltage Compliance:

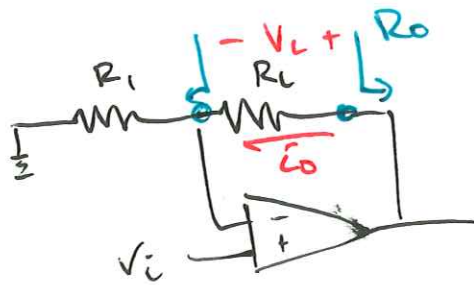
want: $V_{OL} \leq V_o \leq V_{OH}$ **Don't necessarily know V_o !**

what if I only know V_L ?

$$\text{Ex. } V_L = V_o - V_i \rightarrow V_o = V_L + V_i$$

$$V_{OL} \leq V_L + V_i \leq V_{OH}$$

$$V_{OL} - V_i \leq V_L \leq V_{OH} - V_i$$



V-I floating-load converter
Non-Ideal Analysis
 $r_d \neq \infty$, $r_o \neq 0$, $a \neq \infty$

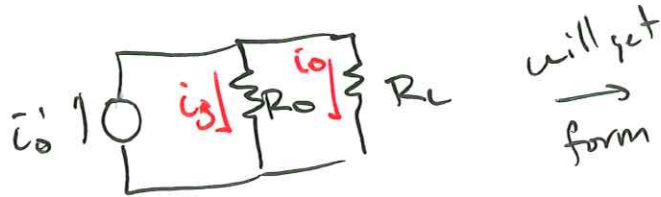
$$\rightarrow i_o = \left[\frac{a - (1+a)(R_i \parallel r_d)/r_d}{r_o + (1+a)(R_i \parallel r_d)} \right] V_i - \frac{V_L}{r_o + (1+a)(R_i \parallel r_d)}$$

$= A V_i$

$\rightarrow$ goes to 0 with $a \rightarrow \infty$

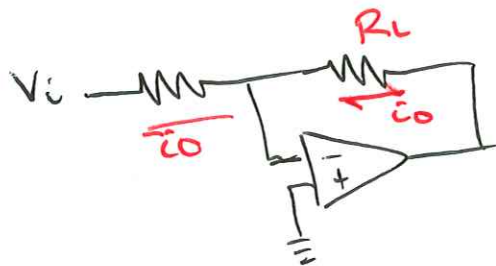
$-\frac{V_L}{R_o}$

~~Ideal~~ current source with load.



will get
form

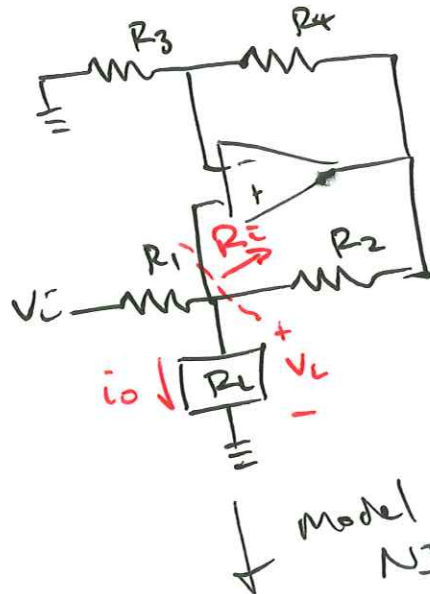
$$i_o = A V_i - \frac{V_L}{R_o}$$



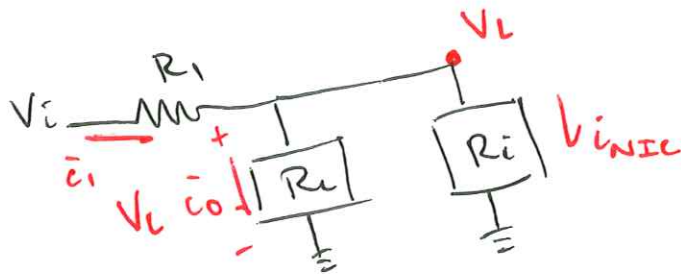
$\rightarrow$ source has to supply i_o !
usually not a good idea!

Grounded Load V-I

Howland - Current Pump



Model with
NIC R_i



from Neg. Imp. Conv. (NIC)

$$R_i = -\frac{R_3}{R_4} R_2$$

After lecture
→ Stability concerns.

KCL:

$$i_i = i_o + i_{NIC}$$

$$\frac{V_i - V_L}{R_1} = i_o + \frac{V_L}{R_i}$$

$$i_o = \frac{V_i}{R_1} - \left(\frac{1}{R_1} - \frac{R_4}{R_2 R_3} \right) V_L$$

$$A_{vi} = \frac{1}{R_1} \quad R_o = \frac{1}{\frac{1}{R_1} - \frac{R_4}{R_2 R_3}}$$

want R_o to be ∞ for ideal current output.

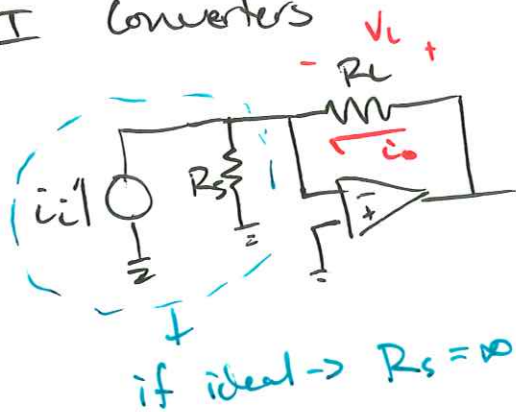
$$\text{want } \frac{1}{R_1} = \frac{R_4}{R_2 R_3} \rightarrow \frac{R_2}{R_1} = \frac{R_4}{R_3}$$

$$\text{let } R_1 = R_2 = R_3 = R.$$

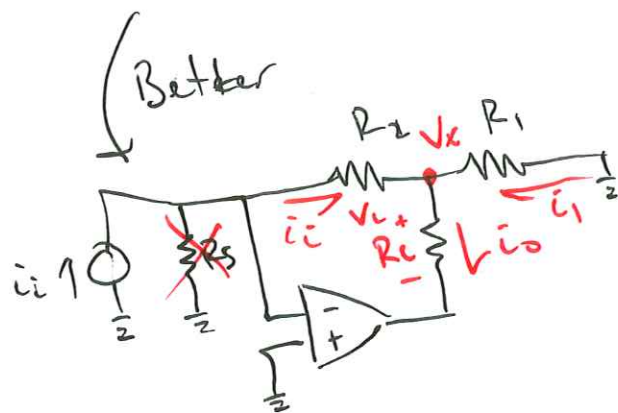
$$\text{then } i_o = \frac{V_i}{R} \rightarrow \frac{i_o}{V_i} = \frac{1}{R}$$

what is i_i ? $R_i = -R$, i_{NIC} is negative, $\rightarrow i_i < i_o$

I-I Converters



→ $i_i = -i_o$
 why not drop the amplifier and
 load R_L directly?



↑
 floating load
 I-I

KCL @ V_x (Ideal Analysis)

$$i_o = i_i + i_1$$

$$V_x = -i_1 R_1 = -i_i R_2$$

$$i_1 = \frac{R_2}{R_1} i_i$$

$$i_o = i_i + \frac{R_2}{R_1} i_i = \left(1 + \frac{R_2}{R_1}\right) i_i$$

A_{vi}

ideal case
 $R_o \rightarrow \infty$

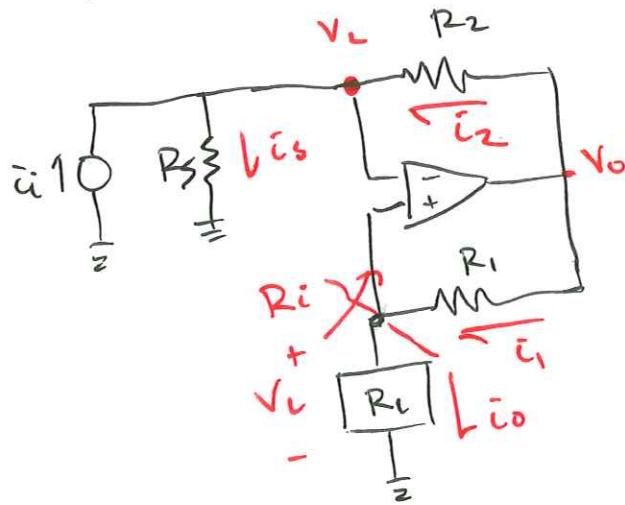
if $R_2 \gg R_1$

then A_{vi} very large.

since R_2 large, i_i is small

(little current from source)

I-I Grounded Load



$$R_i = -\frac{R_1}{R_2} R_s$$

Virtual short:

~~$$\bar{i}_i + \bar{i}_2 = \bar{i}_s = \frac{V_L}{R_s} = \bar{i}_i + \frac{V_o - V_L}{R_2}$$~~

$$\bar{i}_i + \bar{i}_2 = \bar{i}_s = \frac{V_L}{R_s} = \bar{i}_i + \frac{V_o - V_L}{R_2}$$

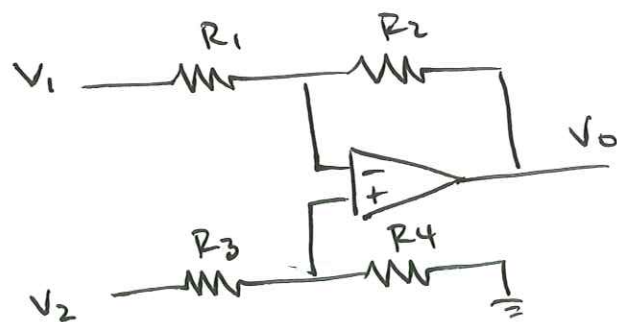
$$\left. \begin{aligned} \bar{i}_2 &= \frac{V_o - V_L}{R_2} \\ \bar{i}_1 &= \bar{i}_o = \frac{V_o - V_L}{R_1} \end{aligned} \right\} \rightarrow \begin{aligned} \bar{i}_2 R_2 &= \bar{i}_o R_1 \\ \bar{i}_o &= \frac{R_2}{R_1} \bar{i}_2 \end{aligned}$$

$$\bar{i}_i + \frac{R_1}{R_2} \bar{i}_o = \frac{V_L}{R_s}$$

$$\bar{i}_o = -\frac{R_2}{R_1} \bar{i}_i + \frac{R_2}{R_1 R_s} V_L$$

$$A_{ii} = -\frac{R_2}{R_1} \quad \hookrightarrow \quad R_o = -\frac{R_1}{R_2} R_s$$

Difference Amplifier



$$\rightarrow V_0 = \frac{R_4}{R_1} \left(\frac{R_1 + R_2}{R_3 + R_4} \right) V_2 - \frac{R_2}{R_1} V_1$$

Typically desired case is:

$$V_0 = A (V_2 - V_1)$$

$$A = \frac{R_2}{R_1}$$

$$R_3 = R_1 \\ R_4 = R_2$$

what if $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ is desired BUT

in actuality: $\frac{R_1}{R_2} = \frac{R_1 + \delta_3}{R_2 + \delta_4} = \frac{R_3}{R_4}$ what happens?

$$\text{let } \frac{R_4}{R_3} = \frac{R_2'}{R_1'} = \frac{R_2}{R_1} (1 - \epsilon)$$

$\epsilon \rightarrow$ some small error value

from ideal equation ...

$$V_0 = \frac{R_2}{R_1} \left[1 - \frac{R_1 + 2R_2}{R_1 + R_2} \frac{\epsilon}{2} \right] (V_2 - V_1) + \frac{R_2}{R_1 + R_2} \epsilon \frac{V_2 + V_1}{2}$$

Diff. mode gain
 A_{dm}

Difference,
 $V_{dm} = \text{"Differential mode" voltage}$

Average.
 $V_{cm} = \text{"Common mode" voltage.}$
common mode gain, A_{cm}

$$V_0 = A_{dm} V_{dm} + A_{cm} V_{cm}$$

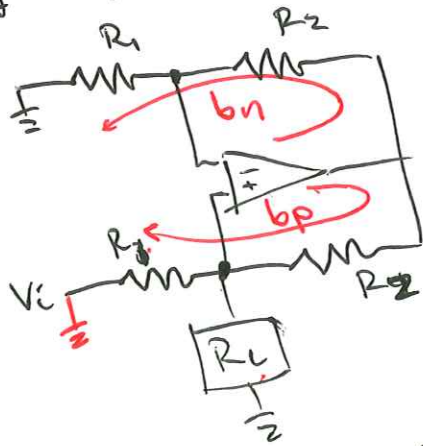
$$V_1 = V_{cm} - V_{dm}/2$$

$$V_2 = V_{cm} + V_{dm}/2$$

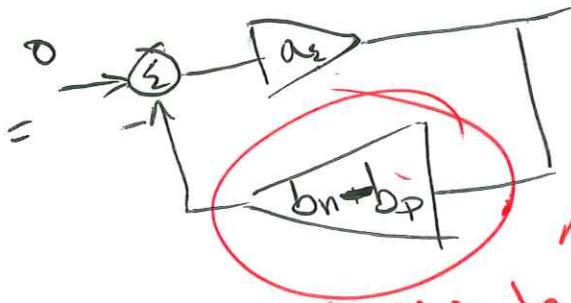
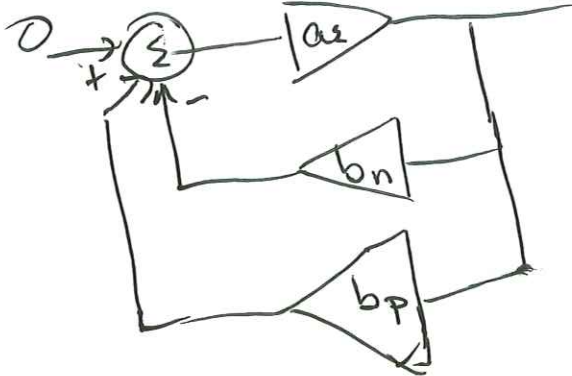
$$V_{dm} = V_2 - V_1$$

$$V_{cm} = \frac{V_2 + V_1}{2}$$

Stability of Howland - Current Pump



$$a_2 = a$$



Need this to be positive!

stability requires negative feedback

don't care about V_i then.

$$\text{Need: } b_n - b_p > 0$$

$$b_n > b_p$$

$$b_n = \frac{V_o}{V_i} = \frac{R_1}{R_1 + R_2}$$

$$b_p = \frac{V_o}{V_i} = \frac{1}{1 + \frac{R_2}{R_L} + \frac{R_2}{R_1}} = \frac{R_1 \parallel R_L}{R_1 \parallel R_L + R_2}$$

$$\frac{R_1}{R_1 + R_2} > \frac{1}{1 + \frac{R_2}{R_L} + \frac{R_2}{R_1}}$$

$$\frac{R_1 + R_2}{R_1} < \frac{1 + \frac{R_2}{R_L} + \frac{R_2}{R_1}}{1}$$

$$1 < 1 + \frac{R_2}{R_L} \rightarrow 0 < \frac{R_2}{R_L}$$

simply: $R_L \neq \infty$